

Lab Report: Home WiFi Network Simulation

Cisco Packet Tracer Implementation & Static IP Provisioning

Objective: Build a wireless star-type topology connecting three host personal computers to a central wireless router in Cisco Packet Tracer, assign static IPv4 addressing parameters, and verify end-to-end connectivity.

1. Topology Construction & File Initialization

The simulation environment was initialized and saved as `HomeWiFiNetwork.pkt`. A wireless router model (WRT300N) was deployed to serve as the central gateway and access point. Three end devices (PCs) were added to the logical workspace, and their default Ethernet network interface cards (NICs) were replaced with wireless expansion modules (WMP300N) to enable wireless association with the router's default SSID.

2. IP Addressing Schema & Subnetting

To ensure proper Layer 3 reachability within the local network, a standard Class C private IPv4 prefix was designated. The wireless router interface was configured with gateway address `192.168.10.1` using subnet mask `255.255.255.0`. Static IP assignments were mapped as follows:

Device Name	IP Address	Subnet Mask	Default Gateway
Wireless Router	192.168.10.1	255.255.255.0	N/A (Gateway)
PC 1 (Client A)	192.168.10.2	255.255.255.0	192.168.10.1
PC 2 (Client B)	192.168.10.3	255.255.255.0	192.168.10.1
PC 3 (Client C)	192.168.10.4	255.255.255.0	192.168.10.1

3. Step-by-Step Configuration Guide

- Open Packet Tracer & Create Project:** Launch Cisco Packet Tracer, create a new file, and save it locally as `HomeWiFiNetwork.pkt`.
- Deploy Wireless Router:** Select *Wireless Devices*, drag the WRT300N router into the workspace canvas.
- Deploy and Modify End Devices:** Place three PCs. Click each PC, open the Physical tab, power off the device, swap out the wired FastEthernet module for the wireless WMP300N module, and power the device back on.

3. Step-by-Step Configuration Guide (Continued)

4. **Configure Static IP Parameters:** Navigate to each PC's *Desktop > IP Configuration* window and enter the static IP, subnet mask, and default gateway as outlined in the schema table above.
5. **Verify Wireless Association:** Check that radio waves link each PC dynamically to the WRT300N wireless access point, verifying Layer 1 and Layer 2 association.

4. Connectivity Testing & Verification

ICMP ping tests were executed from the Command Prompt utility on PC 1 to validate bidirectional transmission across the wireless medium:

```
C:\> ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

C:\> ping 192.168.10.3
Pinging 192.168.10.3 with 32 bytes of data:
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
```

All test pings completed with a 0% packet loss rate, successfully proving that static IP configuration and wireless routing parameters were properly instantiated across the simulated local area network.

5. Summary of Artifacts Generated

The structured lab environment successfully demonstrates local subnet provisioning over wireless infrastructure. All host devices retain active communication paths through the central WRT300N gateway device.